



Formulaic Competence in College-Level Asian English Learner's Argumentative Writing: Examining the Effects of Language Background and Topic

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Abstract The present study examined the effect of language background and topic on productive formulaic competence. Guided by usage-based theory of language learning, this study used a distribution-based approach to the examination of the native-likeness of formulaic usage in English timed argumentative writing. Six indices informed by a large-scale native corpus were chosen to gauge the frequency and association strength of bigrams and trigrams in a total of 778 English timed independent argumentative essays on two topics by 100 native speaker writers, 210 English as a foreign language (EFL) writers and 79 English as a second language (ESL) writers in Asia. Results of a series of linear mixed-effects regression analyses showed that, while EFL writers scored lower on most of the indices than both NS writers and ESL writers, ESL writers did not differ much from NS writers in their use of *n*-grams across topics. Meanwhile, the topic that was deemed more cognitively demanding and of a stronger technical nature elicited more native-like performance in bigram and trigram use across all three language groups. Findings of the study highlight the important effect of input on the development of formulaic competence in a second language, offer empirical support to the use of cognitively complex topics in second language

writing practice, and carry important implications for L2 writing pedagogy.

Keywords Usage-based linguistics · *n*-Gram · Language background · Topic

Introduction

Usage-based linguistics sees language as a large repertoire of symbolic constructions, or form-meaning mappings. One type of construction pervasive in both spoken and written languages (Römer, 2009; Sinclair, 1991) is formulaic units, i.e., (semi-) fixed, recurrent phrases. Formulaic units are believed to be psychologically real as they were found to play an important role in the cognitive processes of language learning and use, including storage, representation, access, etc. (Ellis et al., 2008; Wray, 2002). The appropriate use of formulaic units also helps L2 users signal their membership in a discourse community (Wray, 2002), which is particularly important for L2 writers in academic or professional settings (Hyland, 2008). Despite their importance, formulaic units have been found particularly challenging for L2 learners (Durrant & Schmitt, 2009).

As usage-based theory of language learning posits that the constructions of the target language are learned through “experiencing their exemplars in contextualized usage” (Ellis et al., 2016, p. 42), formulaicity in L2 tends to be defined in terms of its degree of similarity to native norms (Bestgen & Granger, 2014). This means that statistical information on formulaic usage patterns from both L1 and L2 corpus data is essential for the evaluation of L2 formulaic competence. Fortunately, thanks to the easy access to large corpora data, a distribution-based, bottom-up approach to research on formulaic competence, in which the formulaicity

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of a multiword unit is estimated quantitatively via the distribution of the unit in a large-scale native corpus, has recently gained momentum in the field of applied linguistics (Chen, 2019). Studies adopting this approach have helped shed light on the L2 development of formulaic competence (Chen, 2019; Chen & Zhang, 2022; Durrant & Schmitt, 2009; Granger & Bestgen, 2014; Siyanova-Chanturia, 2015; Xia et al., 2022) and on the relationship between productive knowledge of formulaicity and L2 writing quality (Bestgen & Granger, 2014; Bestgen, 2017; Garner et al., 2019, 2020). On the other hand, though usage-based linguistics views input as the primary source for L2 learning (Ellis et al., 2008), and English as a foreign language (EFL) and English as a second language (ESL) learners have long been assumed to differ in the linguistic input available (Granger & Bestgen, 2014), little research has been done on how formulaic competence differs across EFL and ESL essays, which impedes the understanding of the role input plays in the acquisition of L2 productive formulaic knowledge. Meanwhile, an increasing amount of empirical evidence points to a topic effect on the linguistic complexity of L2 writing in that more cognitively demanding topics elicited the use of more complex syntactic structures. However, no empirical research has been conducted to examine the effect that topics with differing cognitive demands may have on the formulaicity in L2 writing, which impedes a fuller understanding of how language production could vary as a function of the cognitive complexity of tasks (Robinson & Gilabert, 2007). Moreover, since none of the extant studies on the differences in formulaic competence across L2 writers and English native speaker (NS) writers (Durrant & Schmitt, 2009; Yoon, 2016) controlled for the topics written on, there may exist a potential confounding effect of topics in their findings.

To address the aforementioned gaps, the present study intended to adopt a distribution-based approach, and examine the use of formulaic units in timed independent argumentative essays by college-level Asian EFL and ESL writers as well as by NS writers across two topics of different cognitive demands.

Literature Review

Operationalization of Formulaic Competence

Adopting a usage-based approach to language learning, the present study operationalized formulaic competence as how native-like writers' use of formulaic units is. The specific type of formulaic units examined is *n*-gram, i.e., a sequence of *n* elements (usually words) that occur directly one after another in a corpus, where *n* is two or more (Gablasova et al., 2017). The collocability of every *n*-gram in a learner's

writing can be assessed based on its distribution in a representative native-speaker corpus. These distributional metrics can then serve as effective measures to assess how native-like each *n*-gram used by learners is (Bestgen, 2017; Bestgen & Granger, 2014; Durrant & Schmitt, 2009).

Two types of corpus-based distributional indices have been used commonly in the operationalization of formulaic competence, i.e., frequency and association strength. Frequency, i.e., how often a formulaic unit appears in a corpus, is an important indicator of the extent to which the use of *n*-grams is native-like. Both longitudinal (Chen & Zhang, 2022; Siyanova-Chanturia, 2015) and cross-sectional studies (Xia et al., 2022) reported an increase of *n*-gram frequency across time or L2 proficiency levels. Garner et al. (2019) also found that higher rated L2 timed independent argumentative essays contained a greater proportion of frequent trigrams from the academic sub-corpus of the Corpus of Contemporary American English (COCA).

The other way of measuring collocational strength is through calculating statistical association. Several measures have been proposed, all of which work on the principle of comparing the number of times a formulaic unit appears in a corpus with the number of times it would be predicted to appear by chance based on the frequency of its component words (Durrant & Schmitt, 2009). Among these measures, Mutual Information (MI), which tends to give higher results on infrequent, strongly associated combinations (Evert, 2009), is one of the most widely used (Gablasova et al., 2017; Gries & Ellis, 2015). A positive relationship between MI and essay scores has been reported (Bestgen, 2017; Bestgen & Granger, 2014; Garner et al., 2019, 2020). Studies that compared association strength cross-sectionally also found an increase in MI from lower L2 proficiency level to higher ones (Granger & Bestgen, 2014; Xia et al., 2022). The findings of longitudinal studies were more varied, with some studies reporting an increase in MI over time (Chen & Zhang, 2022; Siyanova-Chanturia, 2015), and others detecting no difference in this measure (Bestgen & Granger, 2014; Yoon, 2016).

While MI can gauge statistical association, it does not account for the directionality of word combinations (Gries & Ellis, 2015). As association strengths between two words are not symmetrical, it is also necessary to gauge the directional association of word combinations. One commonly used measure is Delta P, a contingency measure from associative learning research. It calculates the probability of an outcome given a cue minus the outcome occurring without the cue present. Chen (2019) and Chen and Zhang (2022) found that L2 learners demonstrated a pronounced growth in forward-directed formulaicity across proficiencies. Garner et al. (2019, 2020) also found that for English argumentative writing, academic bigram Delta P was the strongest predictor of L2 essay quality.

Effects of Language Background and Topic on Formulaic Competence

Usage-based theory of language learning holds that the degree and type of exposure are important determinants of language acquisition (Bestgen & Granger, 2014). Groom (2009) and Waibel (2008) reported that the time spent in an L2 country was a significant predictor of the quality and quantity of L2 formulaic expressions. As language background, i.e., whether one is a native speaker of a language or is learning it as a foreign language or a second language, is closely associated with how much one can be exposed to the target language, it can be hypothesized to have a substantial impact on one's language use. Some studies looked into how NS writers differed from L2 writers in terms of formulaic competence, and found that due to the conservatism in their production, L2 writers tended to use significantly less low-frequency bigrams and strongly-associated bigrams characterized by high MI scores than NS writers in both academic (Durrant & Schmitt, 2009) and argumentative writing (Yoon, 2016).

With regard to ESL and EFL learners' formulaic competence, it was hypothesized that since ESL learners are immersed in the target language, they would perform better than EFL learners, who usually learn the target language in an input-poor environment (Granger & Bestgen, 2014). Yamashita and Jiang (2010) reported that ESL learners made fewer errors and reacted faster than EFL learners in a phrase-acceptability judgment task, and a study on Arabic-speaking learners of English (Alsakran, 2011) found that ESL learners outperformed EFL learners in a test of receptive and productive collocational knowledge. However, no empirical study has probed into the differences in formulaicity in ESL and EFL writing.

Furthermore, extant studies comparing formulaic competence across learners of different language backgrounds tended to view different topics as having a negligible effect on language production. However, as posited by Gablasova et al. (2017), the technical nature of a specific topic can influence the strength of association of formulaic units. For instance, different topics such as social conditioning and global warming in the Trinity Lancaster Corpus of spoken L2 production would lead to the use of pre-modifier-noun pairs of different MI values (Gablasova et al., 2017). So far, no empirical study has been done to verify this hypothesis.

Meanwhile, some recent studies of topic effect on syntactic complexity found that more cognitively demanding topics such as those that inclined toward being public (Atak & Saricaoglu, 2021) and those involving multiple perspective-taking (Sarte & Gnevshva, 2022) led to the use of more complex phrasal modifiers from higher developmental stages (Biber et al., 2011). Since usage-based linguistics sees a less categorical distinction between lexicon and grammar,

it seems worthwhile to investigate if such topic effect also exists in the use of *n*-grams.

In an attempt to address the gaps identified in the literature review above, the present study intended to examine the use of two types of *n*-grams—bigrams, i.e., combinations of two adjacent words, and trigrams, i.e., combinations of three adjacent words, in timed independent argumentative essays by college-level EFL, ESL and NS writers on two topics which differ in cognitive demands and their technical nature. The study chose to focus on bigrams and trigrams, as they were found to be much more prevalent in L2 writing products than longer *n*-grams (i.e., four-grams and five-grams) (Chen, 2019), and were also shown to contribute to L2 writing quality (Garner et al., 2019, 2020).

Specifically, the study seeks to answer the following research questions:

1. How does language background affect bigram and trigram use in timed independent argumentative writing?
2. How does topic affect bigram and trigram use in timed independent argumentative writing?

Methodology

Data

The data examined were extracted from the International Corpus Network of Asian Learners of English (ICNALE) (Ishikawa, 2013). The ICNALE was chosen due to the controls for essay length, topic, task duration, etc., and the inclusion of NS, ESL and EFL essays (see Ishikawa, 2013 for specifics). Based on their scores in standardized English proficiency tests or vocabulary size tests, the L2 writers, who were all college-level learners, were classified into four proficiency bands linked with the European Framework of Reference for Languages (CEFR) scale: A2, B1-1, B1-2, and B2+. The L2 samples used were taken from the ICNALE Edited Essays sub-corpus, which contained 640 essays with ratings. According to Ishikawa (2018), these essays were chosen randomly from the ICNALE Written Essays. Specifically, excluding cases where the number of the original essays was insufficient, twenty essays by writers at each of the four L2 proficiency bands in each of the ten countries and areas in Asia were randomly selected. Furthermore, a total of 62 essays by 31 L2 writers were deleted from this dataset in a list-wise manner, due to one or both essays being off-topic, suspected of plagiarism, or ridden with severe format errors that rendered the essays unparsable, which left the total number of L2 essays to be 578. Two hundred essays written by 100 NS college students were also used in this study, which brings the total number of essays examined to 778. Table 1 presents the descriptive statistics of all 778

Table 1 Descriptive statistics of essays grouped by language background

Group	Topic	No. of texts	No. of words	Word mean	Word SD
EFL	PTJ	210	47,618	226.752	29.211
	SMK	210	46,581	221.814	28.684
ESL	PTJ	79	19,108	241.873	35.610
	SMK	79	18,576	235.139	32.843
NS	PTJ	100	22,224	222.240	22.467
	SMK	100	22,066	220.660	21.718

essays. As revealed by the demographic information of the participants in the supplementary materials, the ESL group was of higher English proficiency than the EFL group. However, since the study intended to present a full picture of the productive formulaic competence of college-level L2 writers in Asia, we opted not to exercise a strict control over the writers' proficiency levels.

The essays in ICNALE were written on two topics: (A) It is important for college students to have a part-time job. (B) Smoking should be completely banned at all the restaurants in the country. Writers should indicate whether they agree or disagree with the statements with reasons and examples. The smoking topic (hereafter referred to as the SMK topic) was a "social topic" (Ishikawa, 2018, p. 127), and was found to involve multiple perspective-taking, whereas the part-time job topic (hereafter referred to as the PTJ topic) was a "personal topic" (Ishikawa, 2018, p. 127) involving a single perspective, as it was mainly argued from the writers' personal perspective or by referring to their own experiences. Since both public topics and topics involving multiple perspective-taking are presumed to be more cognitively demanding (Hamp-Lyons & Mathias, 1994; Robinson & Gilabert, 2007), it is hypothesized that the SMK topic is more cognitively complex than the PTJ topic. Moreover, since the SMK topic has been found to elicit the use of more technical expressions such as "carbon dioxide," "public health," "toxic substances," "lung cancer," etc., it is hypothesized to be of a stronger technical nature.

Text Analysis

Since all writers were required to run spell check using MS word or a similar word processor before completing their writing (Ishikawa, 2013), no text pre-processing was conducted before automated analyses. To examine bigram and trigram use in the writing samples, Tool for the Automatic Analysis of Lexical Sophistication 2.0 (TAALES: Kyle & Crossley, 2015) was used. All TAALES *n*-gram indices are normed by text length (Kyle et al., 2018) and are derived from the BNC (BNC Consortium, 2007), and the COCA (Davies, 2009) as well as its five subsections, i.e., academic,

news, fiction, magazine, spoken, as reference corpora. Since the register under study, i.e., timed independent argumentative writing, is commonly featured in both college-level writing courses and the writing section of academic English tests, such as TOEFL, IELTS, GRE, SAT, etc., we believe the native norm to which the linguistic features in ICNALE essays should be compared ought to be academic in nature. Therefore, the COCA academic sub-corpus, which contains around 88 million words from academic journals published from 1990 to 2012, was chosen as the reference corpus.

Indices Chosen

A total of six indices, including three bigram indices and three trigram indices based on COCA academic, were chosen to gauge the degree of native-likeness of all bigrams and trigrams.

Frequency

A frequency score was assigned to all of the bigrams and trigrams in an essay based on their frequencies in the reference corpus, i.e., COCA academic. Average academic bigram/trigram frequency scores were then calculated for each essay in the corpus.

Association Strength

Since MI was found to violate the assumption of homogeneity of variance, Mutual Information Squared (MI^2), an alternative to MI (Evert, 2009), was used to measure bidirectional association of all bigrams and trigrams. MI^2 of a bigram is the logarithm of the observed co-occurrence of two items (squared) divided by the expected co-occurrence of two items (Evert, 2009). By increasing the influence of the observed *n*-gram frequency as compared to expected frequency, MI^2 counterbalances MI's low-frequency bias. As for trigrams, MI^2 calculates the bidirectional association strength between the first two words and the third word. Average academic bigram/trigram association strength scores were then calculated for each essay in the corpus.

Directional Association Strength

Delta P, which measures the probability of an outcome (O) given a cue (C) minus the probability of an outcome without the cue, was used to gauge the directionality of all bigrams and trigrams. For trigrams, the first two words are the cue and the third word is the outcome. Average academic bigram/trigram directional association strength scores were then calculated for each essay in the corpus.

Statistical Analysis

To examine the effects of language background and topic on the use of bigrams and trigrams, linear mixed-effects models were employed. By relegating the idiosyncrasies of particular writers, topics, etc., to the variances captured by the random effects (Gries, 2015), linear mixed-effects models can provide more accurate estimates of the regression coefficients of the fixed effects than ANOVA. Moreover, this type of statistical analysis is capable of handling unbalanced data (Pinheiro, 2005). Specifically, six linear mixed-effects regression analyses were performed using the lme4 package in R (Version 1.1–26; Bates et al., 2015) with language background (EFL, ESL and NS) and topic (PTJ and SMK) as dummy-coded fixed effects in a 2×3 factorial design. For nested random effects, random intercepts for writers were specified in all models. The best-fitting models were selected using the likelihood ratio test, which was derived from the anova function in Base R. To assess the p -values for fixed effects, significance tests for model coefficients were performed using Satterthwaite approximate degrees of freedom (Luke, 2017). Simple effects and post hoc analyses were performed by comparing estimated marginal means using the emmeans package in R (version 4.1.2, R Core Team, 2021), R Studio IDE (R Studio Team, 2022).

Results

AS shown by the descriptive statistics presented in Table 2, the EFL writers lagged behind both the ESL and the NS writers on the majority of the six indices. The differences between the ESL and the NS writers were smaller compared with those between the EFL and the NS writers. Meanwhile, writers of all three language backgrounds scored higher on all six indices when writing on the SMK topic than on the PTJ topic.

Table 3 presents the results of the six mixed-effects regression analyses. Since topic interacted with language background on bigram frequency ($\text{SMK} \times \text{NS}$: estimate = -63.657 , $p < 0.001$) and trigram MI^2 ($\text{SMK} \times \text{ESL}$: estimate = 0.161 , $p = 0.018$; $\text{SMK} \times \text{NS}$: estimate = 0.140 , $p = 0.026$), simple effects analyses were carried out on these two indices so as to reveal the degree to which one factor is differentially effective at each level of a second factor. Table 4 shows that both the EFL and the ESL writers scored significantly higher on bigram frequency than the NS writers on the SMK topic (EFL-NS: estimate = 35.980 , $p = 0.035$; ESL-NS: estimate = 44.200 , $p = 0.038$), though no significant differences existed among all three groups of writers on the PTJ topic. As shown by Table 5, all three groups were found to score significantly higher on the SMK topic than on the PTJ topic (EFL: estimate = -131.91 , $p < 0.001$; ESL: estimate = -98.99 , $p < 0.001$; NS: estimate = -68.25 , $p < 0.001$). These findings showed that although all three

Table 2 Descriptive statistics of six n -gram measures by topic and language background ($N = 778$)

Structure		Language background		Topic			
		Overall mean	Group mean	PTJ		SMK	
				Mean	SD	Mean	SD
Bigram frequency	205.537	EFL	202.428	136.473	66.443	268.382	125.324
		ESL	241.322	169.390	66.403	276.602	140.808
		NS	198.274	164.148	64.283	232.400	82.796
Bigram MI^2	8.862	EFL	8.792	8.697	0.292	8.888	0.356
		ESL	8.857	8.774	0.239	8.936	0.300
		NS	9.012	8.951	0.221	9.073	0.298
Bigram delta P	0.046	EFL	0.043	0.039	0.009	0.048	0.012
		ESL	0.048	0.046	0.008	0.056	0.013
		NS	0.049	0.045	0.008	0.052	0.011
Trigram frequency	10.208	EFL	9.421	8.744	3.352	10.099	5.168
		ESL	10.238	10.609	4.331	11.924	5.172
		NS	11.023	9.800	3.828	12.246	6.022
Trigram MI^2	8.123	EFL	8.039	8.025	0.438	8.052	0.470
		ESL	8.066	8.101	0.312	8.290	0.387
		NS	8.243	8.159	0.285	8.326	0.335
Trigram delta P	0.005	EFL	0.005	0.004	0.003	0.005	0.004
		ESL	0.005	0.006	0.004	0.007	0.005
		NS	0.006	0.005	0.002	0.007	0.003

Table 3 Summary of the fitted mixed-effects models of n -grams, as predicted by language background and topic ($N=778$)

Measure	Best-fitting model	Fixed effects				Random effects	
		Estimate	SE	t	p	Writer SD	Residual SD
Bigram frequency	Intercept	136.473	6.683	20.422	<0.001***	38.080	89.040
	ESL	32.917	12.781	2.575	0.010*		
	NS	27.674	11.766	2.352	0.019*		
	SMK	131.909	8.689	15.181	<0.001***		
	SMK $\times$ ESL	-24.697	16.620	-1.486	0.138		
	SMK $\times$ NS	-63.657	15.299	-4.161	<0.001***		
Bigram MI ²	Intercept	8.709	0.019	458.799	<0.001***	0.168	0.248
	ESL	0.062	0.032	1.942	0.053		
	NS	0.220	0.030	7.445	<0.001***		
	SMK	0.167	0.018	9.415	<0.001***		
Bigram Delta P	Intercept	0.039	0.001	58.589	<0.001***	0.005	0.009
	ESL	0.007	0.001	6.645	<0.001***		
	NS	0.005	0.001	5.137	<0.001***		
	SMK	0.009	0.001	13.911	<0.001***		
Trigram frequency	Intercept	8.608	0.283	30.389	<0.001***	1.265	4.451
	ESL	1.845	0.448	4.122	<0.001***		
	NS	1.601	0.412	3.886	<0.001***		
	SMK	1.627	0.319	5.099	<0.001***		
Trigram MI ²	Intercept	8.025	0.028	289.179	<0.001***	0.170	0.365
	ESL	0.077	0.053	1.441	0.015		
	NS	0.134	0.049	2.752	0.006**		
	SMK	0.027	0.036	0.769	0.442		
	SMK $\times$ ESL	0.161	0.068	2.372	0.018*		
	SMK $\times$ NS	0.140	0.063	2.228	0.026*		
	Intercept	0.004	0.000	26.499	<0.001***	0.001	0.003
	ESL	0.002	0.000	5.595	<0.001***		
Trigram Delta P	NS	0.001	0.000	3.226	0.001**		
	SMK	0.001	0.000	5.337	<0.001***		

Baselines for categorical variables: Language background = EFL; Topic = PTJ. Random factor: writer.
 *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$

groups of writers used more bigrams of higher frequencies on the SMK topic, the increase was more pronounced in the bigrams used by the two groups of L2 writers than in those used by the NS writers. Figure 1 visualizes the two-way interaction.

As for trigram MI², simple effects analyses (Table 4) showed that there were no significant differences among the three groups of writers on the PTJ topic; however, the EFL writers scored significantly lower on trigram MI² than both the ESL and the NS writers on the SMK topic (EFL-ESL: estimate = -0.238, $p = 0.001$; EFL-NS: estimate = -0.274, $p < 0.001$). As shown in Table 5, though there was a significant increase in trigram MI² in both the ESL and the NS essays across topics, no difference was found across topics in the EFL essays (ESL: estimate = -0.189, $p = 0.019$; NS: estimate = -0.167, $p = 0.020$; EFL: estimate = -0.027, $p = 1.000$). Figure 2 visualizes the two-way interaction.

With regard to the other four indices, i.e., bigram MI², bigram DP, trigram frequency, and trigram DP, no interaction effect was found between topic and language background (see Table 3). As shown in Table 3, significant language background effect was detected, as the EFL writers were found to score significantly lower than the NS writers on all four indices (bigram MI²: estimate = 0.220, $p < 0.001$; bigram DP: estimate = 0.005, $p < 0.001$; trigram frequency: estimate = 1.601, $p < 0.001$; trigram DP: estimate = 0.001, $p = 0.001$), and they also scored significantly lower than the ESL writers on all indices except for bigram MI² (bigram MI²: estimate = 0.062, $p = 0.053$; bigram DP: estimate = 0.007, $p < 0.001$; trigram frequency: estimate = 1.845, $p < 0.001$; trigram DP: estimate = 0.002, $p < 0.001$). Post-hoc analyses (Table 6) showed no significant differences between the ESL writers and the NS writers, except for their scores on bigram MI² (ESL-NS: estimate = -0.158, $p < 0.001$).

Table 4 Summary of simple contrasts between language backgrounds on the two topics

Measures	Topic	Estimate	SE	<i>t</i>	<i>p</i>
Bigram frequency	PTJ				
	EFL–ESL	– 32.920	12.780	– 2.575	0.153
	EFL–NS	– 27.670	11.770	– 2.352	0.284
	ESL–NS	5.240	14.580	0.360	1.000
	SMK				
	EFL–ESL	– 8.220	12.780	– 0.643	1.000
Trigram MI ²	EFL–NS	35.980	11.770	3.058	0.035*
	ESL–NS	44.200	14.580	3.032	0.038*
	PTJ				
	EFL–ESL	– 0.077	0.053	– 1.441	1.000
	EFL–NS	– 0.135	0.049	– 2.752	0.091
	ESL–NS	– 0.058	0.061	– 0.958	1.000
Trigram MI ²	SMK				
	EFL–ESL	– 0.238	0.053	– 4.483	0.001**
	EFL–NS	– 0.274	0.049	– 5.610	<0.001***
	ESL–NS	– 0.036	0.061	– 0.598	1.000

Post-hoc *p* values were Bonferroni corrected. ****p* < 0.001; ***p* < 0.01; **p* < 0.05

Significant topic effect was also found, with the SMK topic eliciting higher scores on all four indices than the PTJ topic (bigram MI²: estimate = 0.167, *p* < 0.001; bigram DP: estimate = 0.009, *p* < 0.001; trigram frequency: estimate = 1.627, *p* < 0.001; trigram DP: estimate = 0.001, *p* < 0.001) (see Table 3).

To sum up, except for bigram frequency on the SMK topic and bigram MI², there was little difference in the ESL and the NS writers' use of bigrams and trigrams across topics.

While the EFL writers scored lower on most of the indices than both the ESL and the NS writers, they surpassed the NS writers in bigram frequency on the SMK topic. Meanwhile, no statistically significant difference was found between the EFL and the ESL writers on either bigram MI² across topics or trigram MI² on the PTJ topic. As for the effect of topics, except for the EFL writer's performance on trigram MI², the SMK topic led to significant increases in all the other five indices across all language groups.

Discussion

The present study examined the use of bigrams and trigrams in a total of 778 essays on two topics, including 578 essays by college-level Asian EFL and ESL writers, as well as 200 essays by college-level NS writers. With regard to language background effect, the EFL writers were found to lag behind both ESL and NS writers in most of the indices measuring the native-likeness of *n*-gram use. A closer look at the two L2 groups showed that those of the highest L2 proficiency level, i.e., the B2+ level, accounted for about 38% and 18.6% of the ESL group and the EFL group, respectively (see supplementary materials for details). A separate series of linear mixed-effects regression analyses on the effect of L2 proficiency and topic on the L2 group's productive formulaic competence found significant differences between B2+ and the other three L2 proficiency levels in bigram Delta P, trigram MI² and trigram Delta P. Such findings aligned with Chen (2019) and Xia et al. (2022), which found formulaic competence generally increased with L2 proficiency. Since all L2 writers were college students, the differences between the EFL and the ESL groups' productive formulaic competence and their L2 proficiency suggested that a more immersion-based L2 exposure facilitated the development of L2 writers' productive formulaic competence (Groom,

Table 5 Summary of simple contrasts between two topics grouped by language background

Measures	Topic	Estimate	SE	<i>t</i>	<i>p</i>
Bigram frequency	EFL				
	PTJ-SMK	– 131.910	8.690	– 15.181	<0.001***
	ESL				
	PTJ-SMK	– 107.210	14.170	– 7.568	<0.001***
Trigram MI ²	NS				
	PTJ-SMK	– 68.250	12.590	– 5.420	<0.001***
	EFL				
	PTJ-SMK	– 0.027	0.036	– 0.769	1.000
	ESL				
	PTJ-SMK	– 0.189	0.058	– 3.254	0.019*
	NS				
	PTJ-SMK	– 0.167	0.052	– 3.238	0.020*

Post-hoc *p* values were Bonferroni corrected. ****p* < 0.001; ***p* < 0.01; **p* < 0.05

Fig. 1 Interaction plot for bigram frequency

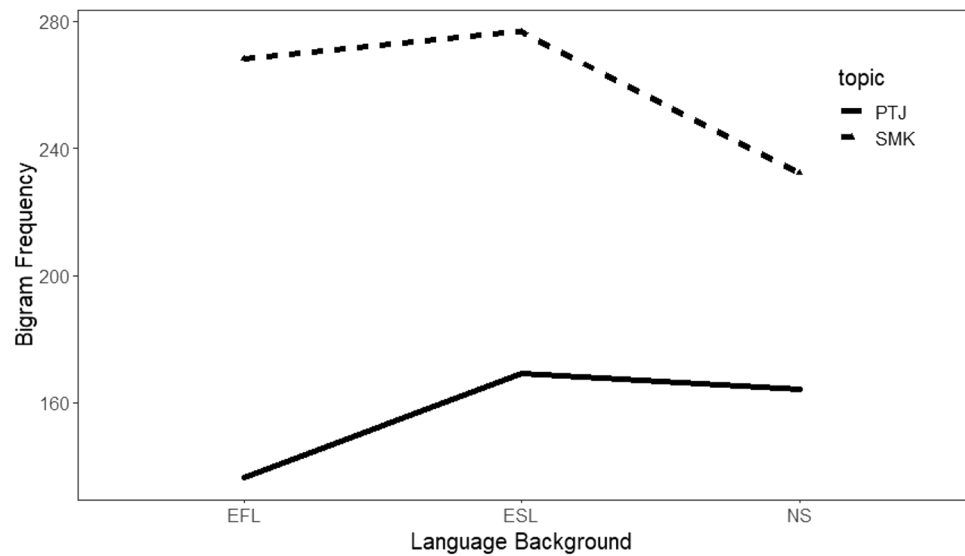
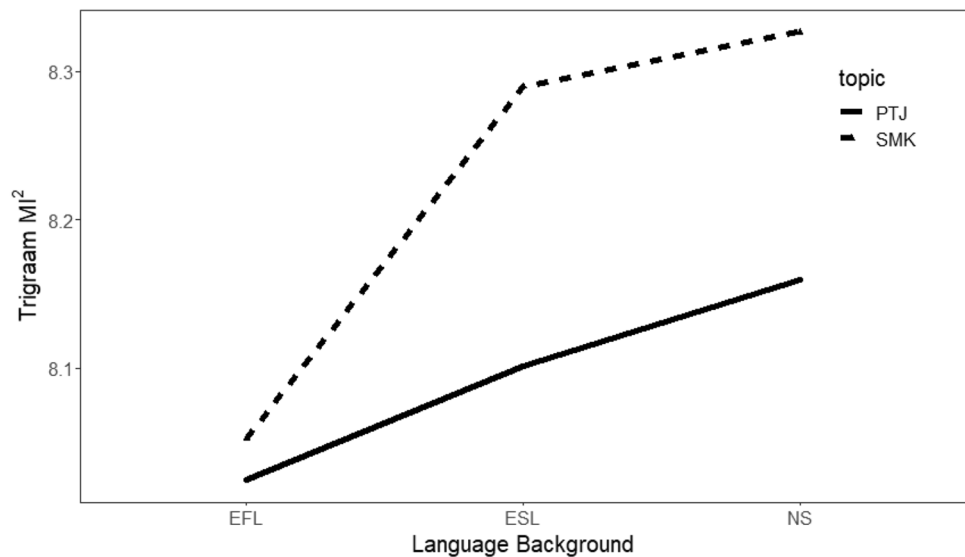


Fig. 2 Interaction plot for trigram MI^2



2009; Waibel, 2008) and their overall L2 competence. On the other hand, the lack of authentic language input in the EFL context may have increased the EFL learners' difficulty in developing formulaic competence and advancing their L2 proficiency. The finding that the EFL writers could not match the ESL writers in bigram DP and all three trigram indices indicated that compared to their ESL counterparts, the EFL learners did not have as good a productive knowledge of directional associations between bigrams and of trigrams common to academic prose. As shown by Fig. 2, while trigram MI^2 in both the ESL and the NS essays increased significantly across topics, its change in the EFL essays across topics was much less evident. Moreover, among the top 20 trigrams with the highest MI^2 values, 13 in the EFL essays were identical across the two topics, while

only six and seven in the ESL and the NS essays were identical across topics respectively (see supplementary materials for the whole list). This finding further illustrated the EFL writers' limited productive knowledge of strongly associated trigrams common to academic prose.

It is also worth noting that though the ESL writers may not have as high an English proficiency level as the NS writers, they did not score significantly lower than the NS writers on the n -gram indices except for bigram MI^2 . They even surpassed the NS writers in bigram frequency across topics. Such findings again illustrated the benefit of immersion-based L2 exposure to the acquisition of L2 productive formulaic knowledge. They also lent support to some psycholinguistic studies that found that native speakers and advanced ESL learners were sensitive to different

Table 6 Summary of post-hoc analyses of pairwise comparisons between language backgrounds

Measures	Estimate	SE	<i>t</i>	<i>p</i>
Bigram MI ²				
EFL–ESL	– 0.062	0.032	– 1.942	0.793
EFL–NS	– 0.220	0.030	– 7.445	<0.001***
ESL–NS	– 0.158	0.037	– 4.306	<0.001***
Bigram Delta P				
EFL–ESL	– 0.007	0.001	– 6.645	<0.001***
EFL–NS	– 0.005	0.001	– 5.137	<0.001***
ESL–NS	0.002	0.001	1.680	1.000
Trigram frequency				
EFL–ESL	– 1.845	0.448	– 4.122	<0.001***
EFL–NS	– 1.601	0.412	– 3.886	0.002**
ESL–NS	0.244	0.511	0.477	1.000
Trigram Delta P				
EFL–ESL	– 0.002	0.000	– 5.595	<0.001***
EFL–NS	– 0.001	0.000	– 3.226	0.020*
ESL–NS	0.001	0.000	2.302	0.328

Post-hoc *p* values were Bonferroni corrected. ****p* < 0.001; ***p* < 0.01; **p* < 0.05

determinants of fluent processing of English—ESL learners to frequency, native speakers to MI (Ellis et al., 2008). Since the EFL learners were also found to score significantly higher on bigram frequency than the NS writers on the SMK topic, and since no significant differences were found between the two L2 groups on bigram MI², it seemed that regardless of their L2 proficiency levels, both groups of L2 learners were more sensitive to frequency than to association strength. On the one hand, this finding lent further support to the effect of the frequency of exposure on L2 learners' acquisition of formulaic sequences (Ellis et al., 2008). Since high-frequency sequences were more likely to be encountered than low-frequency ones (Tomasello & Stahl, 2004), they stood a better chance to be acquired and used by L2 learners. On the other hand, the lack of sensitivity to bigrams of strong association strength marked an important difference between L2 writers and NS writers, suggesting that L2 writers still have a long way to go in their sampling of language.

With regard to topic effect, the study found that the SMK topic, which was hypothesized to be more cognitively complex and of a stronger technical nature, led to increases in all six indices across all language backgrounds. This finding lent support to the Cognition Hypothesis (Robinson & Gilabert, 2007), which posits that a task of higher cognitive complexity promotes higher awareness of the linguistic forms and pushes learners to produce more complex linguistic structures to meet the functional and communicative demands of the task. A closer look at the bigrams and trigrams extracted

showed that the SMK topic led to the use of more complex linguistic structures of strong statistical association than the PTJ topic. Such structures included nouns with prepositional phrases as postmodifier, such as “backlash against (MI² = 10.910),” “opportunities for (MI² = 11.539),” “relationship between (MI² = 14.590),” “large proportion of (MI² = 10.610),” “the results of (MI² = 11.032),” “different kinds of (MI² = 11.481),” and noun complement clauses such as “the fact that (MI² = 11.951),” “no doubt that (MI² = 12.066),” etc. These structures are considered complex as they are in the later stages of the hypothesized developmental trajectory of complexity features (Biber et al., 2011). Meanwhile, since these nominal structures are also a defining feature of written academic texts (Biber et al., 2011), the increased use of them could help enhance the native-likeness of the written products.

This finding also provided empirical evidence for Gablasova et al. (2017)'s hypothesis that “the technical nature of a specific topic can influence the strength of collocations as measured by the MI-score” (p. 174). The present study found that the SMK topic, which is about banning smoking in all restaurants, did lead to the use of quite a few technical terms such as “public health (MI² = 12.662),” “lung cancer (MI² = 12.901),” “blood pressure (MI² = 13.166),” “toxic substances (MI² = 13.868),” “global warming (MI² = 14.578),” “carbon monoxide (MI² = 14.962),” “carbon dioxide (MI² = 16.208),” etc., while the only technical term found on the PTJ topic, which is about doing a part-time job, is “higher education (MI² = 13.305)”.

Conclusion

Findings of the study carry important theoretical and pedagogical implications. Theoretically, the language background effect discovered offers empirical support to the importance of input in second language acquisition. Specifically, the EFL group's restricted access to authentic language may have further hampered the development of their formulaic competence and their overall English proficiency. Both L2 groups' sensitivity to high-frequency *n*-grams also implies that due to a greater possibility of being exposed to such formulaic units, these structures can be more easily acquired by L2 learners. The topic effect found lends support to not only the Cognition Hypothesis (Robinson & Gilabert, 2007), which argues that tasks of higher cognitive complexity lead to more complex language use, but also Gablasova et al.'s (2017) argument that topics of a stronger technical nature could affect the association strength of formulaic units.

Pedagogically, first, since both L2 groups had difficulty mastering bigrams with strong association strength, and since such formulaic units tend to fulfill distinctive functions, carry specific meanings and have the grammatical

well-formedness as complete phrases (Ellis et al., 2008), explicit instruction should be provided in L2 classes to help students notice and acquire them. Second, freely available web-based corpora of native academic prose can be used to supplement textbooks, so as to provide EFL learners with more authentic language input. Third, as EFL learners seemed particularly weak in their command of trigrams, special attention should be given to these structures in class instruction. Finally, to help elicit more complex and native-like language use, topics that are more cognitively complex and of a stronger technical nature can be used in writing practices, at least for learners of or above intermediate English proficiency level.

The study has several limitations, which can be addressed in future studies. First, as the three groups of writers differed in their English proficiency, some of the differences observed may have arisen from their distinct English proficiency level rather than their nativeness status. To better untangle English proficiency and language background effect, future studies should compare NS, ESL and EFL writers of similar proficiency levels. Second, as all L2 learners examined are from Asia, findings of the study may not be generalized to L2 population in other regions. More studies on L2 learners from other countries and places should be done to verify the differences observed in this study. Third, the study focused on bigrams and trigrams. However, since a wide range of structures, such as collocations, lexical bundles, collocations have been identified as formulaic units, future studies should examine a wider variety of formulaic structures so as to enable a fuller understanding of formulaic competence in L2 writing. Fourth, writers' perception of the differences in the two topics were predicted a priori, rather than empirically established. Future studies could try using questionnaires and interviews to gather evidence regarding both L1 and L2 writers' perception of topic differences.

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